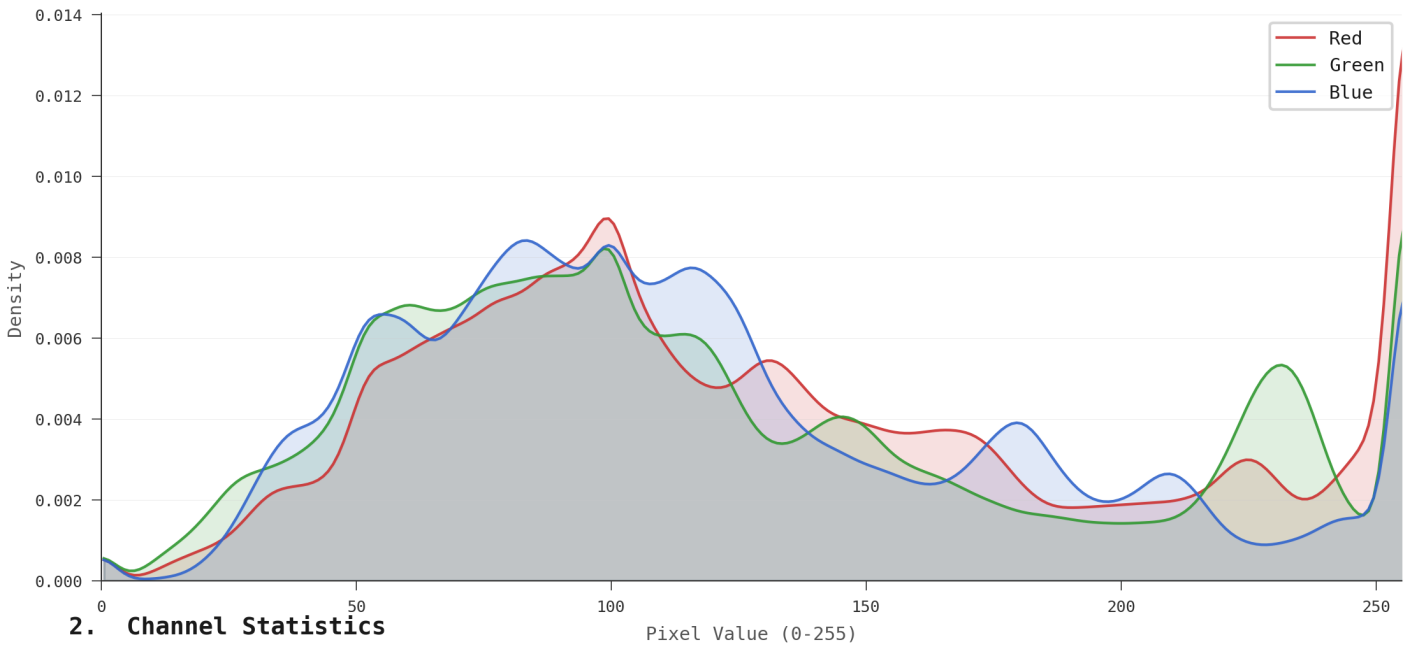
Reference Data Sheet: KODIM08

Subject: Seifenfabrikation Street Buildings

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	127.9621	63.7385	4062.5961	-0.7074	0.5369	0	255
Green	121.0786	65.2947	4263.4034	-0.7235	0.5882	0	255
Blue	115.9959	57.3863	3293.1915	-0.2341	0.6954	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9665	0.9162
G	0.9665	1.0000	0.9746
B	0.9162	0.9746	1.0000

Pairwise Pearson Correlation Coefficients

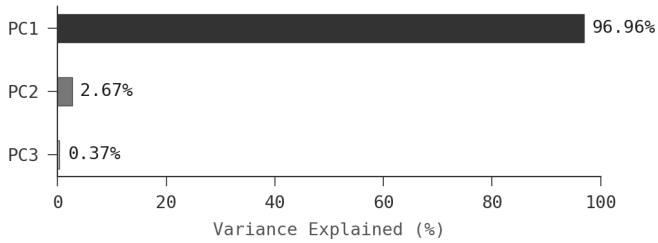
r(R-G)	=	**0.966460**
r(R-B)	=	**0.916208**
r(G-B)	=	**0.974616**
Avg |r|	=	**0.952428**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{4062.6065} & 4022.2107 & 3351.2393 \\
4022.2107 & \mathbf{4263.4142} & 3651.9205 \\
3351.2393 & 3651.9205 & \mathbf{3293.1999}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM08

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5879	**-0.6130**	-0.5279	11265.8584
PC2	-0.7265	0.1130	0.6778	310.1984
PC3	0.3558	-0.7820	0.5118	43.1638

6. Derived Metrics

Condition Number

261.00

(l1/l3) PC1/PC2: 36.32 | PC2/PC3: 7.19

Blue Channel Independence

4.7%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6130 | Green dominant

Classification: Strongly One-Dimensional (PC1 = 96.96%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.886158	0.920963	0.903561
Green	0.888479	0.923422	0.905951
Blue	0.862235	0.906698	0.884466

8. Average Local Variance (3x3)

Channel	Value
Red	635.3855
Green	656.8125
Blue	605.4027

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 261.00 | PC1: 96.96%

Correlation: G–B pair dominates at r = 0.974616 | R–G: 0.966460 | R–B: 0.916208

Blue Channel: 4.7% independent | 95.3% captured by PC1

Spatial: Avg autocorrelation: 0.897993 | Avg local variance: 632.5336

Eigenvector: Green dominant | G (-0.6130) leads | R (-0.5879), B (-0.5279)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."